

NATQ-002 — benchmark freeze report

100 accepted cases · 136 gold spans · 28 rejected on premise · 0 validator failures · head `03061c6` · 2026-09-04

Read the refusals, not just the acceptances. 100 cases were accepted, but 28 of 150 authored questions were *rejected because the frozen corpus never states their premise*, and three adjudications were caught wrong and corrected after the fact. Those two numbers, not the headline, are what decide whether this benchmark can be trusted — a set built by admitting every question would score systems on evidence that does not exist.

100 ACCEPTED CASES	136 GOLD SPANS	28 REJECTED ON PREMISE	0 RESIDUAL COLLISIONS	0 INTEGRITY FAILURES
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1 — The pipeline, 150 to 100

Stage	Cases	What happened
Authored	150	Five isolated cold-context agents, <code>tool_uses=0</code> , blind to corpus, evidence and each other's questions
Adjudicated	150	Every question checked against the frozen corpus
Supported	122	Gold evidence located and validated
Rejected	28	Premise not documented — no gold span can exist
Eligible	112	After 10 cases dropped by the collision rulings
Accepted	100	Selected under a rule fixed before selection

By authoring slice

Slice	Authored	Supported	Rejected	Accepted
A	30	23	7	20
B	30	21	9	19
C	30	25	5	20
D	30	27	3	21
E	30	26	4	20

Slice B yielded only 19 eligible cases, so an even 20-per-slice split was impossible. B contributes all 19 and the shortfall goes to D. Early in the build slice A alone suggested a 27% rejection rate; the true rate across all five slices is 28/150, which is why no yield decision was taken on a partial sample.

2 — What was refused, and why that matters

A rejection here is not a failure to search. It is a finding that the corpus never states the question's premise, so admitting the case would train the benchmark to reward retrieving a passage that does not answer it. Every rejection records the probes run and, where adjacent text exists, why anchoring there would be a false positive.

Case	Question	Why refused
A01	how do I stop claude from calling the same tool over and over	Premise not documented in the frozen corpus.
A06	migrating from functions/function_call to tools/tool_choice, what actu	Premise not documented in the frozen corpus.
A08	getting an error that my schema root must be an object	Premise not documented in the frozen corpus.
A11	claude keeps saying "I'll use the tool" in text before actually callin	REJECT_ON_PREMISE / UNSUPPORTED_SCOPE, per coordinator ruling.
A17	pydantic model as the response format, does it work with optional fiel	Premise not documented in the frozen corpus.
A18	what's the max number of tools i can define	Premise not documented in the frozen corpus.
A29	we call it function calling internally but the docs say tool use, are	Premise not documented in the frozen corpus.
B01	why does my stream just stop halfway through with no error	Premise not documented in the frozen corpus.
B06	I need to show tokens as they arrive in a react app, what's the least	Premise not documented in the frozen corpus.
B08	what happens if the client disconnects mid-stream — do I still get bil	Premise not documented in the frozen corpus.

28 rejections in total; ten shown.

3 — Evidence collisions

Different questions can resolve to the same gold evidence. Left unchecked, that double-weights one retrieval capability. All 9 exact-span groups were ruled under a single stated rule: **keep the case whose gold answer requires the largest portion of the shared span**, ties by earlier case ID. Dropped cases are not deleted — their packets keep full evidence and carry the ruling.

Group	Kept	Reason
A09, A10	A09	A09's gold answer exercises the whole span: the default value false AND both consequence branches (at most one under auto, exactly one under any/tool)
A07, D26	A07	A07's answer is the contrast the two spans state. D26 asks when to choose, which requires selection criteria the spans do not give and the packet had
A26, C20	C20	C20 is answered in full by the span. A26 is compound — temperature 0 AND forced tool use — and the span addresses only the temperature half.
C07, E30	E30	E30 asks for the breakpoint placement, which is the span's literal content. C07 wraps the same rule in a cost goal the span does not mention.
C09, E10	C09	C09 applies the hierarchy rule to the exact edit it describes. E10 requires first inferring which level changed, a step outside the span.
C22, C23	C23	C23 matches the span's scope — Claude models and thinking. C22 was authored as an OpenAI question and needed provider re-scoping to fit the same evide
B23, B24, D06	B23	B23's answer enumerates the full retryable set, using the whole span. B24 needs only 'it retries' and D06 only 'twice'.
B30, E16	E16	E16 uses both clauses of the span — HTTP-level and enqueued-request-level limits. B30 asks whether budgets are shared, which the span does not state.
E02, E13	E02	E02's answer is the formula the span states outright. E13 additionally infers a threshold from the downscale rule, a step beyond the spans.

The rule you choose decides whether the benchmark clears 100.

Under the governing exact-span rule, 112 unique cases remain from 122 supported — comfortably above target. Group the same cases by the `cs_v1_control` chunk their spans fall in and only 92 remain, which is **below** 100. A chunk is the unit a retrieval system is actually scored on, so this is a real design choice, not bookkeeping. It is recorded and left to the coordinator; no case was excluded for chunk overlap alone.

Selection used chunk diversity only as a tiebreak within the governing rule, which cut chunk-level overlap from 20 groups covering 50 cases across all supported cases to **13 groups covering 27 cases** in the accepted 100, with **0 residual exact-span collisions**.

4 — Selection rule, fixed before selecting

- Eligible pool = SUPPORTED minus cases dropped by COLLISION-RULINGS-001.
- Quota 20 per slice; a slice with fewer than 20 eligible contributes all of them and the shortfall is reallocated to the slice with the largest remaining surplus.
- Within a slice, greedily take the case whose gold chunk set overlaps least with chunks already accepted; ties by case ID ascending.
- No retrieval or reranking system consulted. Chunk membership read from the frozen chunk table.

Slice	A	B	C	D	E
Eligible	21	19	23	25	24
Quota	20	19	20	21	20

5 — Errors caught, by assertion rather than by judgement

Three adjudications were wrong and were corrected. B22 was rejected with the reason “no document in the snapshot states this” — written after searching a single document. The string was in the corpus. That error triggered a corpus-wide audit of every rejection literal, which found two more: A15 was a case-sensitivity miss (searched `retina`, corpus writes `macOS Retina displays`) and E27 was scoped to the wrong documents. All three are now supported.

Eight questions carried truncated text. The freeze asserts every accepted question equals its authored text byte-for-byte, and it refused to run. An audit found eight packets carrying question strings truncated by console listings rather than read from the authored file (A22, A29, B02, B08, C15, D29, E23, E30). Two had lost the second half of the asker's framing. All restored; adjudications, spans and claims were unaffected.

The rule adopted and applied to every later rejection: **per-document absence is not snapshot absence**, and substring searches must be case-checked.

6 — Integrity of the frozen set

- every gold span re-read from the frozen corpus reproduces its quote exactly
- every critical string is present in its own evidence
- every provider matches the provider of its source documents
- every accepted question equals its authored text byte-for-byte

136 gold spans re-read from the frozen corpus, 0 failures. 33 of the 100 accepted cases carry more than one gold span; 1 case cites evidence spanning two providers (A02), carried on span-level version_ids.

Quantity		Value
Corpus snapshot		snap_689e336380a054d8039dc35b2c09cd0a
Documents in snapshot / referenced by gold		202 / 42
Provider split		anthropic 74, cross-provider 1, openai 25
benchmark_file_sha256	1450b9e1ba892d84d2ecab279cfa229b128a02108a4704b56393382b6c5bf741	
canonical_content_sha256	d1f2b51f61a05c7652cd420c32bedb6b72b2ad8cfe14a43ab0fe3d49774c9d60	

The canonical hash covers case IDs, question text and span offsets only, so it survives cosmetic edits to answers or notes and breaks if any question or gold span changes.

7 — Still gated

Gate	State
split decided	False
holdout locked	False
preregistered	False

Constraints held

Constraint	State
natq002 scored by any system	False
natq001 holdout opened	False
exp029a fabricated	False
question wording changed	False
cases deleted	False
collisions resolved using system output	False

The validation/reserve split is deliberately not decided here. It determines which cases become exposed development data and which stay pristine, and it must not be chosen by looking at system results. It returns to the coordinator along with the collision-rule question in section 3.